ARJUNA JEE 2027

Lecture -11 | Mathematics

TRIGONOMETRIC FUNCTIONS

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Topics *to be covered*

A **Trigonometric Series**

B **Question Practice**

C

D





LAST CLASS RECAP

Answer the following Questions:

1. $\sin 2A + \sin 2B + \sin 2C =$ $4 \sin A \sin B \sin C$

2. $\cos 2A + \cos 2B + \cos 2C =$ $-1 - 4 \cos A \cos B \cos C$

3. $\sin A + \sin B + \sin C =$ $4 \cos\frac{A}{2} \cos\frac{B}{2} \cos\frac{C}{2}$

4. $\cos A + \cos B + \cos C =$ $1 + 4 \sin\frac{A}{2} \sin\frac{B}{2} \sin\frac{C}{2}$

5. $\tan(nA) + \tan(nB) + \tan(nC) =$ $\tan(nA) \cdot \tan(nB) \cdot \tan(nC)$

6. $\cot A \cot B + \cot B \cot C + \cot C \cot A =$ 1.

7. $\tan\frac{A}{2}\tan\frac{B}{2} + \tan\frac{B}{2}\tan\frac{C}{2} + \tan\frac{C}{2}\tan\frac{A}{2} =$ 1.

H.W. Discussion

Question [JEE Mains-2022]

$2\sin\left(\frac{\pi}{22}\right)\sin\left(\frac{3\pi}{22}\right)\sin\left(\frac{5\pi}{22}\right)\sin\left(\frac{7\pi}{22}\right)\sin\left(\frac{9\pi}{22}\right)$ is equal to

A $\frac{3}{16}$

B $\frac{1}{16}$

C $\frac{1}{32}$

D $\frac{9}{32}$

$$2\cos\left(\frac{10\pi}{22}\right)\cdot\cos\left(\frac{8\pi}{22}\right)\cdot\cos\left(\frac{6\pi}{22}\right)\cdot\cos\frac{4\pi}{22}\cdot\cos\frac{2\pi}{22}$$

$$2\cos\frac{\overset{0}{2\pi}}{22}\cdot\cos\frac{\overset{20}{4\pi}}{22}\cdot\cos\frac{\overset{40}{8\pi}}{22}\left(+\cos\frac{\overset{80}{16\pi}}{22}\right)\left(+\cos\frac{\overset{160}{32\pi}}{22}\right)$$

$$\frac{2\sin\left(\frac{64\pi}{22}\right)}{2^5\sin\left(\frac{2\pi}{22}\right)} = \frac{1}{2^4} = \frac{1}{16}$$

$\frac{\pi}{2}-\frac{\pi}{22}=\frac{10\pi}{22}$

$\sin\frac{\pi}{22}=\cos\left(\frac{10\pi}{22}\right)$

$\cos\theta = -\cos(\pi+\theta)$

$\cos\frac{10\pi}{22} = -\cos\left(\pi+\frac{10\pi}{22}\right)$

$= -\cos\frac{32\pi}{22}$

Question

83%

If $A+B+C=\pi$, then the value of expression $\frac{\cos A}{\sin B \sin C}+\frac{\cos B}{\sin C \sin A}+\frac{\cos C}{\sin A \sin B}$ is

Youtube Method

put $A=B=C=60^\circ$ ✓

$$\frac{2\cos A\sin A+2\cos B\sin B+2\cos C\sin C}{2\sin A\sin B\sin C}$$

$$\frac{\sin 2A+\sin 2B+\sin 2C}{2\sin A\sin B\sin C}$$

$$\frac{4\sin A\sin B\sin C}{2\sin A\sin B\sin C}=2$$ Ans

Question

If $\cos^2 A + \cos^2 B + \cos^2 C = 1$, then $\triangle ABC$ is

A. Equilateral

B. Isosceles

C. Right angled

D. None of these

$\cos^2 A + \cos^2 B = \sin^2 C$.

$\cos^2 A + 1 - \sin^2 B = \sin^2 C$.

$\cos^2 A - \sin^2 B = \sin^2 C - 1$

$\cos(A+B)\cos(A-B) = +\cos^2 C$.

$+\cos C \mid \cos C \{\cos(A-B) - \cos C\} = 0$.

OR

$\frac{1+\cos 2A}{2} + \frac{1+\cos 2B}{2} + \frac{1+\cos 2C}{2} = 1$

$2 + c2A + c2B + c2C = 2$

$-1 - 4\,cA\,cB\,cC = -1$

$-4\,cA\,cB\,cC = 0$.

Question

$$\# \quad \frac{x}{y-x} = \frac{1}{\left(\frac{y-x}{x}\right)}$$

Let A, B, C be the three angles such that $A + B + C = \pi$. If $\tan A \cdot \tan B = 2$, then find the value of $\frac{\cos A \cos B}{\cos C}$.

$\cos(C) = \cos(\pi - (A+B))$

$= -\cos(A+B)$

$\frac{\cos A \times \cos B}{-\cos(A+B)} = \frac{CA\,CB}{SA\,SB - CA\,CB} = \frac{1}{\left(\frac{SA\,SB}{CA\,CB} - \frac{CA\,CB}{CA\,CB}\right)}$

$= \frac{1}{\tan A \tan B - 1}$

$= \frac{1}{2-1}$

$= 1$

$-(CA\,CB - SA\,SB)$

Question

RAM

In $\triangle ABC$, if $\cot A + \cot B + \cot C = 0$ then find the value of $\cos A \cos B \cos C$. = -1

$\cot A + \cot B = -\cot C$.

$\frac{\cos A}{\sin A} + \frac{\cos B}{\sin B} = -\frac{\cos C}{\sin C}$

$\frac{\cos A \sin B + \sin A \cos B}{\sin A \sin B} = -\frac{\cos C}{\sin C}$

$\sin(A+B) = \sin(\pi - C) = \sin C$.

$\frac{\sin C}{\sin A \sin B} = -\frac{\cos C}{\sin C}$ — (1)

$\frac{\sin B}{\sin A \sin C} = -\frac{\cos B}{\sin B}$ — (2)

$\frac{\sin A}{\sin B \sin C} = -\frac{\cos A}{\sin A}$ — (3)

(1) × (2) × (3) ⇒

$1 = \frac{\sin A \sin B \sin C}{(\sin A \sin B \sin C)^2} = -\frac{\cos A \cos B \cos C}{(\sin A \sin B \sin C)}$

Acidic Question

If $x + y + z = xyz$, then show that

$$\frac{3x - x^3}{1 - 3x^2} + \frac{3y - y^3}{1 - 3y^2} + \frac{3z - z^3}{1 - 3z^2} = \left(\frac{3x - x^3}{1 - 3x^2}\right)\left(\frac{3y - y^3}{1 - 3y^2}\right)\left(\frac{3z - z^3}{1 - 3z^2}\right)$$

HHP.

put

$x = \tan A$

$y = \tan B$

$z = \tan C$

$\tan A + \tan B + \tan C = \tan A \tan B \tan C$

$\Downarrow$

$A + B + C = \pi$

$\tan 3A + \tan 3B + \tan 3C = \tan 3A \cdot \tan 3B \cdot \tan 3C$

$$\tan 3A = \frac{3\tan A - \tan^3 A}{1 - 3\tan^2 A}$$

$$= \frac{3x - x^3}{1 - 3x^2}$$

TRIANGLE INEQUALITIES

I. For ΔABC : $1 < \cos A + \cos B + \cos C \leq \frac{3}{2}$

Also, $0 < \sin\frac{A}{2} \sin\frac{B}{2} \sin\frac{C}{2} \leq \frac{1}{8}$

II. For ΔABC : $0 < \sin A + \sin B + \sin C \leq \frac{3\sqrt{3}}{2}$

Also, $0 < \cos\frac{A}{2} \cos\frac{B}{2} \cos\frac{C}{2} \leq \frac{3\sqrt{3}}{8}$

III. For Acute angle ΔABC : $\tan A + \tan B + \tan C \geq 3\sqrt{3}$

IV. For ΔABC : $-1 < \cos A \cos B \cos C \leq \frac{1}{8}$

Also, $0 < \sin A \sin B \sin C \leq \frac{3\sqrt{3}}{8}$

(iii) for acute angle

$\tan A + \tan B + \tan C = \tan A \times \tan B \times \tan C$

$\tan A, \tan B, \tan C \rightarrow$ +ve

$\tan A, \tan B, \tan C \rightarrow$ AM $\geq$ GM

$$\frac{\tan A + \tan B + \tan C}{3} \geq (\tan A \times \tan B \times \tan C)^{\frac{1}{3}}$$

$$\frac{x}{3} \geq (x)^{\frac{1}{3}}$$

cube

$$\frac{x^3}{27} \geq x^2 \Rightarrow x^2 \geq 27$$

$$x \geq 3\sqrt{3}$$

(I) "ΔABC"

$$\cos A + \cos B + \cos C = 2\cos\frac{A+B}{2}\cos\left(\frac{A-B}{2}\right) + 1 - 2\sin^2\frac{C}{2}$$

$\cos\frac{A+B}{2} = \sin\frac{C}{2}$

$$2\sin\frac{C}{2}\cos\left(\frac{A-B}{2}\right) + 1 - 2\sin^2\frac{C}{2} \le 2\sin\frac{C}{2} + 1 - 2\sin^2\frac{C}{2}$$

$\#\ \max. = 1 \ (A = B)$

$$2\sin\frac{C}{2}\cos\left(\frac{A-B}{2}\right) + 1 - 2\sin^2\frac{C}{2} \le \frac{3}{2} - 2\left(\sin\frac{C}{2} - \frac{1}{2}\right)^2$$

$\left(\sin\frac{C}{2} - \frac{1}{2}\right)^2 \ge 0$

$\max. = \frac{3}{2}, \ \sin\frac{C}{2} = \frac{1}{2}$

$1 - 2(\alpha^2 - \alpha)$

$\Downarrow$

$1 - 2\left(\left(\alpha - \frac{1}{2}\right)^2 - \frac{1}{4}\right)$

$\frac{3}{2} - 2\left(\alpha - \frac{1}{2}\right)^2$

TRIANGLE INEQUALITIES

(याद करते है) **

I. For ΔABC : $1 < \cos A + \cos B + \cos C \leq \frac{3}{2}$

Also, $0 < \sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2} \leq \frac{1}{8}$

II. For ΔABC : $0 < \sin A + \sin B + \sin C \leq \frac{3\sqrt{3}}{2}$

Also, $0 < \cos\frac{A}{2}\cos\frac{B}{2}\cos\frac{C}{2} \leq \frac{3\sqrt{3}}{8}$

III. For Acute angle ΔABC : $\tan A + \tan B + \tan C \geq 3\sqrt{3}$.

IV. For ΔABC : $-1 < \cos A \cos B \cos C \leq \frac{1}{8}$

Also, $0 < \sin A \sin B \sin C \leq \frac{3\sqrt{3}}{8}$

for min. value in case of (i), (ii), (iv)

$A \simeq 179.999999°$

$B \& C \simeq 0.0000001°$

TRIGONOMETRIC SERIES

RAR

Type -01 : SINE & COSINE series with angles in A.P.

$$\sin(\alpha) + \sin(\alpha+\beta) + \sin(\alpha+2\beta) + \ldots\ldots + \sin(\alpha+(n-1)\beta) = \frac{\sin\left(\frac{n\beta}{2}\right)\sin\left(\alpha+(n-1)\frac{\beta}{2}\right)}{\sin\frac{\beta}{2}}$$

α = first angle, β = common-diff of angles, n = Total Terms.

$$\cos(\alpha) + \cos(\alpha+\beta) + \cos(\alpha+2\beta) + \ldots\ldots + \cos(\alpha+(n-1)\beta) = \frac{\sin\left(\frac{n\beta}{2}\right)\cos\left(\alpha+(n-1)\frac{\beta}{2}\right)}{\sin\left(\frac{\beta}{2}\right)}$$

PROOF

$$\sin(\alpha) + \sin(\alpha+\beta) + \sin(\alpha+2\beta) + \ldots\ldots + \sin(\alpha+(n-1)\beta) = \frac{\sin\left(\frac{n\beta}{2}\right)\sin\left(\alpha+(n-1)\frac{\beta}{2}\right)}{\sin\frac{\beta}{2}}$$

$$T_1 = \frac{2\sin\overset{A}{\alpha}\cdot\sin\overset{B}{\frac{\beta}{2}}}{2\sin\frac{\beta}{2}} = \frac{1}{2\sin\frac{\beta}{2}}\left\{\cos\left(\alpha-\frac{\beta}{2}\right) - \cos\left(\alpha+\frac{\beta}{2}\right)\right\}$$

$$T_2 = \frac{2\sin\overset{A}{(\alpha+\beta)}\sin\overset{B}{\frac{\beta}{2}}}{2\sin\frac{\beta}{2}} = \frac{1}{2\sin\frac{\beta}{2}}\left\{\cos\left(\alpha+\frac{\beta}{2}\right) - \cos\left(\alpha+\frac{3\beta}{2}\right)\right\}$$

$$= \frac{1}{2\sin\frac{\beta}{2}}\left\{\cos\left(\alpha+\frac{3\beta}{2}\right) - \cos\left(\alpha+\frac{5\beta}{2}\right)\right\}$$

$$\vdots$$

Ques:-

① $\sin 5^\circ + \sin 12^\circ + \sin 19^\circ + \sin 26^\circ + \sin 33^\circ = \dfrac{\sin\left(\eta\frac{\beta}{2}\right)\sin\left(\alpha+(\eta-1)\frac{\beta}{2}\right)}{\sin\left(\frac{\beta}{2}\right)}$

$\alpha = 5^\circ, \quad \beta = 7^\circ, \quad \eta = 5$

$= \dfrac{\sin\left(5\times\frac{7^\circ}{2}\right)\sin\left(5^\circ+4\times\frac{7^\circ}{2}\right)}{\sin\left(\frac{7^\circ}{2}\right)}$

Ans

RAPID FIRE

Value of $\cos\frac{\pi}{19}+\cos\frac{3\pi}{19}+\cos\frac{5\pi}{19}+\ldots\ldots+\cos\frac{17\pi}{19}=$

$$=\frac{\sin\left(9\times\frac{\pi}{19}\right)\cos\left(\frac{\pi}{19}+8\times\frac{\pi}{19}\right)}{\sin\left(\frac{\pi}{19}\right)}$$

($\frac{\beta}{2}$ and $\frac{\beta}{2}$ marked)

$\alpha=\frac{\pi}{19}$, $\beta=\frac{2\pi}{19}$, $n=9$.

$\frac{\beta}{2}=\frac{\pi}{19}$.

$$=\frac{2\sin\frac{9\pi}{19}\cdot\cos\frac{9\pi}{19}}{2\sin\frac{\pi}{19}}$$

$$=\frac{\sin\left(\frac{18\pi}{19}\right)}{2\sin\frac{\pi}{19}}=\frac{1}{2}$$

$$\#\ \sin\left(\frac{18\pi}{19}\right)=\sin\left(\pi-\frac{\pi}{19}\right)$$

$$=\sin\frac{\pi}{19}.$$

Acidic Question

$[141.4] = 141$

The value of $[100(x-1)]$ is where $[x]$ is the greatest integer less than or equal to x and $x = \dfrac{\sum_{n=1}^{44^\circ} \cos n^\circ}{\sum_{n=1}^{44^\circ} \sin n^\circ} = \dfrac{\cos 1^\circ + \cos 2^\circ + \cos 3^\circ + \ldots + \cos 44^\circ}{\sin 1^\circ + \sin 2^\circ + \sin 3^\circ + \ldots + \sin 44^\circ}$

A 140

B 141

C 142

D 144

$x = \cot\left(\dfrac{45^\circ}{2}\right) = \cot(22.5^\circ) = \dfrac{1}{\tan(22.5^\circ)} = \dfrac{1}{\sqrt{2}-1} \times \dfrac{\sqrt{2}+1}{\sqrt{2}+1} = \sqrt{2}+1$

$22.5^\circ = \dfrac{\pi}{8}$

$x = \sqrt{2}+1$

$x-1 = \sqrt{2}$

$100(x-1) = 100\sqrt{2} = 141.4$ (using $\sqrt{2} = 1.414$)

$$= \frac{\cos 1^\circ + \cos 2^\circ + \cos 3^\circ + \ldots + \cos 44^\circ}{\sin 1^\circ + \sin 2^\circ + \sin 3^\circ + \ldots + \sin 44^\circ}$$

$$N^r = \cos 1^\circ + \cos 2^\circ + \cos 3^\circ + \ldots + \cos 44^\circ = \frac{\sin\left(44 \times \frac{1^\circ}{2}\right) \cos\left(1^\circ + 43 \times \frac{1^\circ}{2}\right)}{\sin\left(\frac{1^\circ}{2}\right)}$$

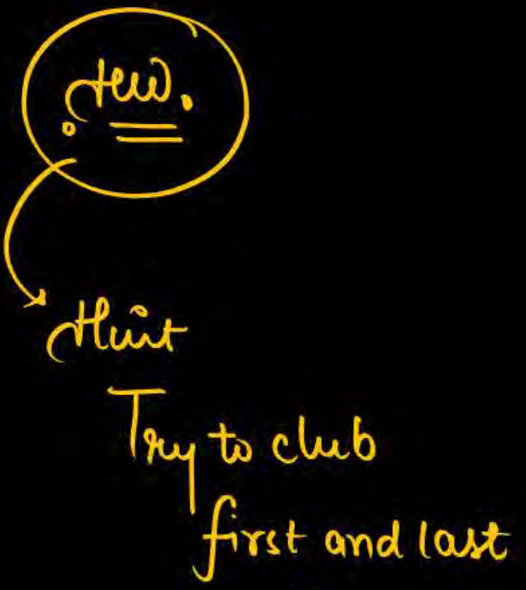
$\alpha = 1^\circ,\ \beta = 1^\circ,\ \eta = 44$

$$D^r = \sin 1^\circ + \sin 2^\circ + \sin 3^\circ + \ldots + \sin 44^\circ = \frac{\sin\left(44 \times \frac{1^\circ}{2}\right) \sin\left(1^\circ + 43 \times \frac{1^\circ}{2}\right)}{\sin\left(\frac{1^\circ}{2}\right)}$$

$$= \cot\left(\frac{45^\circ}{2}\right)$$

Challenger Question

Let $S = \sum_{r=1}^{90} (2r)\sin(2r)^\circ$ then value of $S/90$ is (are) :



A $\cot 1^\circ$

B $\tan 1^\circ$

C $\dfrac{\cos^2 20^\circ - \sin^2 19^\circ}{\sin^2 26^\circ - \sin^2 25}$

D $\sqrt{\dfrac{1-\cos 2^\circ}{1+\cos 2^\circ}}$

Question

$\sum_{r=1}^{n} \sin^2 \frac{r\pi}{n}$ is equal to $S = \sin^{②} \frac{\pi}{n} + \sin^{②} \frac{2\pi}{n} + \sin^{②} \frac{3\pi}{n} + \cdots\cdots + \sin^{②} \frac{n\pi}{n}$

A $\frac{n+1}{2}$

B $\frac{n-1}{2}$

C $\frac{n}{2} - 1$

D $\frac{n}{2}$

$$S = \sin^2\frac{\pi}{n} + \sin^2\frac{2\pi}{n} + \sin^2\frac{3\pi}{n} + \cdots\cdots + \sin^2\frac{n\pi}{n}$$

$$\sin^2\theta = \frac{1-\cos 2\theta}{2} \rightarrow = \frac{1}{2} - \frac{1}{2}\cos 2\theta$$

$$S = \frac{1}{2} - \frac{1}{2}\cos\frac{2\pi}{n} + \frac{1}{2} - \frac{1}{2}\cos\frac{4\pi}{n} + \frac{1}{2} - \frac{1}{2}\cos\frac{6\pi}{n} + \cdots\cdots$$

$$S = n \times \frac{1}{2} - \frac{1}{2}\left\{\cos\frac{2\pi}{n} + \cos\frac{4\pi}{n} + \cos\frac{6\pi}{n} \cdots\cdots\right\}$$

$\alpha = \frac{2\pi}{n}$, $\beta = \frac{2\pi}{n} \rightarrow \frac{\beta}{2} = \frac{\pi}{n}$, (n)

$$S = \frac{n}{2} - \frac{\sin n\left(\frac{\pi}{n}\right)\cos\left(\frac{2\pi}{n} + (n-1)\frac{\pi}{n}\right)}{\sin\left(\frac{\pi}{n}\right)}$$

($\sin n\left(\frac{\pi}{n}\right) \rightarrow 0$)

Find the sum of series: $S = \cos^3\theta + \cos^3 2\theta + \cos^3 3\theta + \ldots$ upto n-terms

$$S = \left(\frac{1}{4}\cos 3\theta + \frac{3}{4}\cos\theta\right) + \left(\frac{1}{4}\cos 6\theta + \frac{3}{4}\cos 2\theta\right) + \left(\frac{1}{4}\cos 9\theta + \frac{3}{4}\cos 3\theta\right) + \ldots$$

$$\frac{1}{4}\{\cos 3\theta + \cos 6\theta + \cos 9\theta + \ldots\} + \frac{3}{4}\{\cos\theta + \cos 2\theta + \cos 3\theta + \ldots\}$$

$\alpha = 3\theta$

$\beta = 3\theta$

(n)

$$\cos 3\theta = 4\cos^3\theta - 3\cos\theta$$

$$\frac{1}{4}\cos 3\theta + \frac{3}{4}\cos\theta = \cos^3\theta$$

Find the sum of series: $S = \cos^3\theta + \cos^3 2\theta + \cos^3 3\theta + \ldots$ upto n-terms

Ans: $S = \frac{1}{4}\left(\frac{\sin\left(\frac{3n\theta}{2}\right)\cos\left(\frac{3(n+1)\theta}{2}\right)}{\sin\left(\frac{3\theta}{2}\right)} + 3\frac{\sin\left(\frac{n\theta}{2}\right)\cos\left(\frac{(n+1)\theta}{2}\right)}{\sin\left(\frac{\theta}{2}\right)}\right)$

DIBY !!

RAPID FIRE DIBY'S

DIBY-A

The expression $\frac{\cos^3\alpha-\cos 3\alpha}{\cos\alpha}+\frac{\sin^3\alpha+\sin 3\alpha}{\sin\alpha}$ is equal to:

(A) 1 (B) 2 (C) 3 (D) 4

[Ans. : C]

DIBY-B

Express $\cos 5A$ in terms of $\cos A$

(A) $16\cos^5 A-20\cos^3 A+5\cos A$ (B) $16\cos^5 A-12\cos^3 A+3\cos A$

(C) $16\cos^5 A-20\cos^3 A+3\cos A$ (D) $32\cos^2 A-32\cos^3 A+6\cos A$

[Ans. : A]

DIBY-C

The value of the product $\cos 36^\circ.\cos 72^\circ$ is equal to :

(A) $\frac{1}{2}$ (B) $\frac{1}{4}$ (C) $\frac{1}{8}$ (D) $\frac{\sqrt{5}}{4}$

[Ans. : B]

RAPID FIRE DIBY'S

DIBY-D

Show that: (i) $\sin 105^\circ + \cos 105^\circ = \cos 45^\circ$

(ii) $\sin 75^\circ - \sin 15^\circ = \cos 105^\circ + \cos 15^\circ$

(iii) $\sin((n+1)A)\sin((n-1)A) + \cos((n+1)A)\cos((n-1)A) = \cos 2A$

DIBY-E

Let $S = \sin\sqrt{2} - \sin\sqrt{3}$ and $C = \cos\sqrt{2} - \cos\sqrt{3}$ then which one is correct?

(A) $S > 0$ and $C > 0$

(B) $S > 0$ and $C < 0$

(C) $S < 0$ and $C > 0$

(D) $S < 0$ and $C < 0$

[Ans. : A]

DIBY-F

The expression $\frac{\sec 8A - 1}{\sec 4A - 1}$ is equivalent to :

(A) $\frac{\tan 8A}{\tan 4A}$

(B) $\frac{\tan 8A}{\tan 2A}$

(C) $\frac{\tan 4A}{\tan -2A}$

(D) $\frac{\tan 2A}{\tan -8A}$

[Ans. : B]

RAPID FIRE DIBY'S

DIBY-G

What is the exact value of $\cos^2 48^\circ - \sin^2 12^\circ$

(A) $\frac{\sqrt{5}-1}{8}$ (B) $\frac{\sqrt{5}+1}{8}$ (C) $\frac{\sqrt{5}+1}{4}$ (D) $\frac{\sqrt{5}-1}{4}$

[Ans. : B]

DIBY-H

What is the exact value of $4\cos\frac{\pi}{10} - 3\sec\frac{\pi}{10} - 2\tan\frac{\pi}{10}$

(A) 0 (B) 1 (C) $\sqrt{5}$ (D) $\frac{1}{2}$

[Ans. : A]

DIBY-I

The value of the expression $\frac{\sin 50^\circ + \sin 100^\circ + \sin 210^\circ}{\sin 25^\circ \sin 50^\circ \sin 105^\circ}$ is equal to:

(A) 1 (B) 2 (C) 3 (D) 4

[Ans. : D]



Homework

Re-attempt all Class Examples

DPP & DIBY

It is never about end result

It's always about Journey

Thank You

#futureIITians